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Current status of cardiology fellow training based on COCATS-4 framework: A cross-sectional study from three tertiary care institutes across India

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ABSTRACT

Background: The Core Cardiology Training Symposium (COCATS) guidelines, endorsed by the American College of Cardiology (ACC) emphasize competency-based training and establish milestones to be met during cardiology training. However, India lacks such guidelines for training cardiology fellows.
Objectives: The study aimed to evaluate the self-declared competencies of cardiology fellows across multiple preselected domains as per the COCATS-4 recommendations.
Methods: The study was conducted at three premier institutions across India between June 2022 and December 2024. A total of 52 cardiology fellows who had recently completed their 3-year core cardiology training course were included in the study. These candidates completed a questionnaire that included a checklist assessing different levels of competency in various preselected aspects of cardiovascular medicine. Candidates were also asked about their awareness of COCATS-4 standards and cardiology training in India.
Results: While all trainees reported Level 1 competency in cardiac catheterization, heart failure, and adult congenital heart disease, a significantly lower percentage of trainees achieved Level 1 competency in nuclear medicine (1.9 %), cardiac MRI (3.8 %), and cardiac CT (17.3 %). At Level 2, cardiac catheterization (73.1 %) and echocardiography (63.5 %) had the highest self-declared competencies, whereas nuclear medicine (3.8 %), cardiac MRI (0 %), and cardiac CT (0 %) had the lowest. Only 23.1 % of trainees were aware of COCATS-4, and 82.7 % believed that standardized guidelines would improve cardiology training in India.
Conclusions: The results emphasize the wide variability in self-reported competencies of cardiology trainees across India and highlight the deficiencies in certain core areas like cardiovascular imaging.

1. Introduction

The core cardiology training in India began more than half a century ago. Over the years, there has been a gradual increase in the number of both public and private institutions providing training in the field of core cardiology as DM or DNB. After completing training, these medical professionals are considered eligible and competent to provide independent clinical care to patients with cardiovascular diseases. Although these training courses are designed to impart adequate knowledge and skills to the trainees, there is a lack of standardization in India.[1–3]

On the contrary, the United States has a rigorous system of regularly updated recommendations in various core areas of cardiovascular medicine. The fourth version of Core Cardiology Training Statement (COCATS) is the present iteration. They were initially published in 1995 as recommendations for training in adult cardiovascular medicine.[4] The recommendations were gradually updated to include peripheral interventions and cardiac MRI in the second iteration, and electrophysiology and cardiac CT were added in the third iteration.[5,6] Recently, there has been an emphasis on competency-based training. The objective is to assess competency across core areas in cardiology and establish

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milestones to be met as the fellows progress in training (COCATS-4).[7,8] The fourth iteration focuses especially on research during training, the use of multimodality imaging, and critical care cardiology. COCATS 4 has divided training into three levels (level I to III). The emphasis of COCATS-4 is on Level I competency. Level I competency are those skills and knowledge that all cardiology fellows must attain during a standard 3-year training program. Level II competency is the one which is achieved when a fellow spends 6 months or more during the 3-year training on a specific aspect of cardiology and is objectively evaluated. Level III competency are those skills and knowledge that requires additional training beyond the standard 3 years training course (Objective competency criteria within each domain has been described in Supplementary material 1).

The first step in developing similar recommendations for cardiology training in India is to assess the existing level of competence based on the established COCATS framework. Understanding the self-perceived competencies of cardiology trainees can offer valuable insights into the strengths and gaps in current training programs. This cross-sectional observational study was conducted to evaluate the self-declared competencies of cardiology trainees across multiple preselected domains, as outlined in the COCATS-4 recommendations. It is the first study to assess the strengths and gaps in the competencies achieved in cardiology training in India.

2. Methods

The study was conducted at three premier institutions across India: the Postgraduate Institute of Medical Education and Research (PGIMER), Chandigarh; the All India Institute of Medical Sciences (AIIMS), New Delhi; and Kasturba Medical College (KMC), Manipal. The responses were received from eligible candidates between June 2022 and December 2024. The institutional ethics committee of all three institutes approved the study, with PGIMER serving as the coordinating institute (Ref. No. IEC/2021/1726). A total of 52 candidates who had completed their 3-year core cardiology training course (DM Cardiology) were included in the study. After informed and written consent, these candidates filled out a questionnaire that contained a checklist for different levels of competency in various preselected aspects of cardiovascular medicine, i.e. electrocardiography, echocardiography, nuclear medicine, cardiac computed tomography, cardiac MRI, peripheral artery disease, cardiac catheterization, electrophysiology, heart failure, critical care and adult congenital heart disease (Supplementary material 1).

The questionnaire was prepared per the COCATS-4 framework with prior permission. Level I competency has been described objectively in the COCATS -4 guidelines for electrocardiography, echocardiography, nuclear medicine, computed tomography, cardiac MRI, peripheral artery disease, cardiac catheterization, electrophysiology, heart failure, critical care cardiology and adult congenital heart disease. Level II competency has been described objectively for echocardiography, nuclear medicine, computed tomography, cardiac MRI, peripheral artery disease, cardiac catheterization, electrophysiology and heart failure. Level III competency has been objectively described only for echocardiography, cardiac catheterization and adult congenital heart disease. These categories, which were objectively described at all three levels, were included in the questionnaire (Supplementary Material 1). The responses were recorded as yes (score of 1) or no (score of zero). The number of yes responses for each question was tabulated for analysis. Candidates were also asked three questions about their awareness of COCATS-4 standards and cardiology training in India (Supplementary Material 1).

3. Results

A total of 52 eligible candidates participated in the study, comprising 29 from PGIMER, 14 from KMC, and 9 from AIIMS. Candidates fulfilling all the criteria for a particular level of competency in a category were considered to possess corresponding "self-declared complete competency". The number of candidates deemed to possess self-declared complete competency is reported in Table 1 and depicted in Fig. 1, categorized by the level of competency and core area of cardiology training involved. All of the candidates ($n = 52$, 100 %) felt they had achieved complete level I competency in cardiac catheterization, heart failure and adult congenital heart disease. The categories with the least self-declared level I competency were nuclear medicine, cardiac MRI and computed tomography, with only 1 (1.9 %), 2 (3.8 %) and 9 (17.3 %) fellows declaring competence, respectively. For level II competency, categories with a maximum number of self-declared complete competencies were cardiac catheterization ($n = 38$, 73.1 %) followed by echocardiography ($n = 33$, 63.5 %). Like level I, the lowest number of self-declared complete competencies was seen in computed tomography ($n = 0$), cardiac MRI (0) and nuclear medicine ($n = 2$, 3.8 %).

The mean scores in each category are shown in Table 1. The results of the mean competency score align with the 'number of candidates with complete competencies in each category, as reported earlier. Only 12 candidates (23.1 %) were aware of COCATS-4 guidelines for cardiology training, while 24 candidates (46.2 %) were unaware of any national guidelines for cardiology training. Of the 52 candidates, 43 (82.7 %) believed introducing curriculum guidelines, such as COCATS-4, would improve cardiology training.

Table 1
Number of candidates who carry self-declared complete competencies in each category and the mean scores.

	No. of candidates who carry self-declared complete competency in each category (%), Total = 52	Mean (±SD) scores in different categories (Range: 0–1)
Level I Competency		
Electrocardiography-Level I	48 (92.3 %)	0.98 ± 0.06
Echocardiography-Level I	37(71.2 %)	0.89 ± 0.17
Nuclear Cardiology- Level I	1(1.9 %)	0.03 ± 0.15
Cardiac computed tomography-Level I	9(17.3 %)	0.36 ± 0.37
Cardiac MRI-Level I	2(3.8 %)	0.32 ± 0.28
Peripheral artery disease-Level I	34(65.4 %)	0.72 ± 0.41
Cardiac catheterization-Level I	52(100 %)	1 ± 0
Electrophysiology-Level I	28(53.8 %)	0.85 ± 0.17
Heart failure-Level I	52(100 %)	1 ± 0
Critical Care-Level I	30(57.7 %)	0.78 ± 0.27
Adult cyanotic congenital heart disease-Level I	52(100 %)	1 ± 0
Level II Competency		
Echocardiography-Level II	33(63.5 %)	0.80 ± 0.30
Nuclear Cardiology- Level II	2(3.8 %)	0.17 ± 0.25
Cardiac Computed tomography-Level II	0	0.12 ± 0.69
Cardiac MRI-Level II	0	0.02 ± 0.09
Peripheral artery disease-Level II	16(30.8 %)	0.58 ± 0.33
Cardiac catheterization-Level II	38(73.1 %)	0.83 ± 0.32
Electrophysiology-Level II	4(7.7 %)	0.40 ± 0.31
Heart failure-Level II	25(48.1 %)	0.48 ± 0.50
Level III competency		
Echocardiography-Level III	23(44.2 %)	0.66 ± 0.33
Cardiac catheterization-Level III	31(59.6 %)	0.59 ± 0.49
Adult cyanotic congenital heart disease-Level III	40(76.9 %)	0.76 ± 0.42

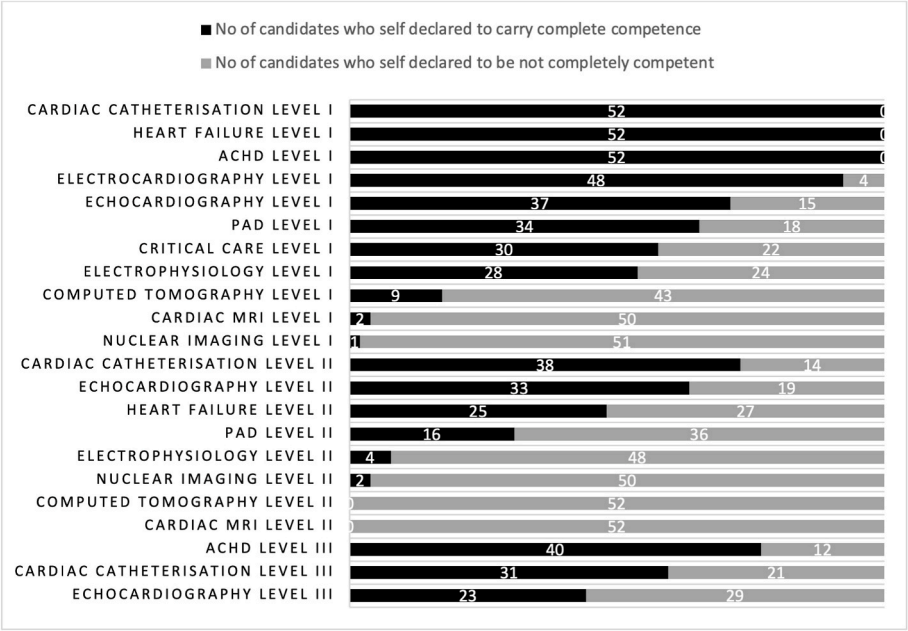

Fig. 1. Horizontal bar chart showing the number of candidates who self-declared their competence levels across various domains of cardiology training. Blue bars indicate the number of candidates who declared themselves to have complete competence. Grey bars indicate the number of candidates who self-declared to be not completely competent.

4. Discussion

The results of this survey highlight significant variability in self-declared competency levels across different domains of cardiology training in India. While all participants (100 %) reported Level I competency in cardiac catheterization, heart failure, and adult congenital heart disease, considerably lower Level 1 competency rates were observed in nuclear medicine (1.9 %), cardiac MRI (3.8 %), and cardiac CT (17.3 %). For Level 2 competency, cardiac catheterization (73.1 %) and echocardiography (63.5 %) were the most commonly achieved skills, whereas only 3.8 % reported Level 2 competency in nuclear medicine and none in cardiac CT and CMR. The trend persisted for Level 3 competencies. These results align with existing literature emphasising the predominance of procedural training in traditional modalities over advanced imaging techniques.[9,10] There are subtle variations in the results among the participants from the three institutes.

It is important to emphasize that the COCATS guidelines were implemented to ensure Level 1 competence across all core areas of cardiology training to ensure that fellows were sufficiently skilled to manage patients at the end of the training period independently.[11] In this regard, it is particularly concerning that core areas of cardiology training, especially those related to imaging modalities, have very low levels of self-declared Level I competence. The importance of imaging modalities has undeniably increased over the past few decades, with recent guidelines championing imaging as Class I recommendations for several indications, including chronic coronary syndromes and management of valvular and congenital heart disease.[12–14] Similarly, the emphasis on critical care cardiology training is justified in the latest guidelines because adequate training can improve the likelihood of good outcomes in these sicker patients.[15] Ultimately, these guidelines aim to improve patient care; therefore, efforts must be made to ensure their implementation to the greatest extent possible.

Several factors may have contributed to these disparate results across different domains. One major factor is the accessibility of training resources. Cardiac catheterization and echocardiography are fundamental procedures in cardiology, with well-established infrastructure, experienced faculty, and adequate patient exposure available in most cardiology training programs. In contrast, advanced imaging modalities like cardiac MRI, nuclear cardiology, and cardiac CT are recent and dynamic additions that have not had adequate penetration into training programs and curricula.[16,17] Moreover, CT and MRI-based imaging have traditionally been the primary focus in the training of radiology residents. Cardiology trainees have not been given adequate time, opportunities, and responsibilities during their core 3-year training to improve their imaging skills, despite the availability of CT/MRI in the vast majority of medical institutions that impart cardiology training in India. Furthermore, faculty expertise plays a crucial role in shaping the competencies of trainees. Institutions with dedicated specialists in advanced imaging techniques are more likely to offer structured training and mentorship in these fields. In centres where such expertise is limited, fellows may not receive adequate guidance or opportunities to develop competency in these subspecialties. A rotation through various concerned departments for the fellows will be beneficial, rather than relying solely on the expertise of individual faculty for all matters.

Notably, only 23.1 % of respondents were aware of the COCATS-4 training guidelines, and 46.2 % acknowledged the lack of national cardiology training standards in India. However, 82.7 % of participants agreed that the introduction of structured training guidelines similar to COCATS-4 would improve cardiology training. These findings underscore the need for standardized competency frameworks to bridge training gaps and ensure uniform expertise across institutions. Like the COCATS-4 guidelines, national guidelines formulated by the National Medical Commission (NMC) must rely on the six pillars of cardiology training: medical knowledge, patient care, interpersonal and communication skills, professionalism, practice-based learning, and systems-based practice.[18,19] Task forces should be formed with experts in various cardiology subspecialties, guiding the development of guidelines and curriculum with a special emphasis on endemic diseases such as rheumatic heart disease and Eisenmenger syndrome. Policy changes by the NMC, which mandate the availability of imaging modalities in all institutions providing cardiology training and ensuring that fellows in training receive adequate exposure, will go a long way in improving competency levels in these categories. Periodic assessments in core areas of cardiology training will foster competence and learning among the fellows. The NMC should administer a standardized, objective exam at the end of training for licensing and certification purposes, ensuring

uniformity. All medical institutions providing cardiology training should be required to regularly update the progress and assessment of trainees to the NMC. The NMC website (https://www.nmc.org.in/information-desk/for-colleges/pg-curricula-2/) for training does not have a separate curriculum-based guideline for cardiology, whereas such guidelines have been established for most super-specialities, including neurology, gastroenterology, and rheumatology. Bidirectional feedback, where trainees and faculty mutually evaluate and provide constructive feedback to one another, will enhance and accelerate competency-based training.[20–22]

5. Limitations

Although this is a novel study, only 52 participants could be included. The three institutes participating in the study are centres of excellence in India, with high patient volumes and access to the latest technologies and advancements. Our results are not generalizable to other Indian institutes because of a marked disparity in the training curriculum across India. This study was based on self-reported competency levels subject to certain inherent biases. Further research involving a larger cohort from multiple centres might provide deeper insights into training needs and inform policy recommendations.

6. Conclusion

This study underscores the variability in self-declared competency levels among cardiology trainees across different domains, highlighting significant gaps in advanced imaging training. The findings emphasize the urgent need for standardized competency-based guidelines to ensure comprehensive cardiology education and uniform skill development across institutions in India.

Data availability

All the data relevant to this study are available with the corresponding authors and can be provided at reasonable request.

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None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ihj.2025.05.010.

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